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Shock, Hemodynamics, Sepsis & Cardiac ICU Care

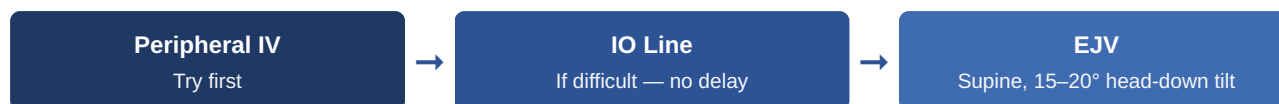
Classify shock and initiate fluid/vasoactive therapy safely; recognize sepsis and anaphylaxis early; apply the 4Ds/ROSE framework to fluid therapy; manage hemodynamic instability in the post-operative cardiac surgical child.

3. Shock, Hemodynamics, Sepsis & Cardiac ICU Care

Learning Objectives

- Classify shock by type and severity and secure timely vascular access (peripheral, IO, EJV).
- Run the 1-hour sepsis bundle and recognise anaphylaxis, giving IM adrenaline without delay.
- Apply the 4Ds/ROSE framework to fluid resuscitation, maintenance, and de-escalation.
- Recognise fluid overload and know when to restrict or actively remove fluid.
- Recognise low cardiac output syndrome and pulmonary hypertensive crisis in the post-cardiac-surgery child.

1. Vascular Access



Give adequate analgesia unless comatose. EJV avoided in: neck trauma/infection, thrombosis/vascular anomaly, unclear cervical spine.

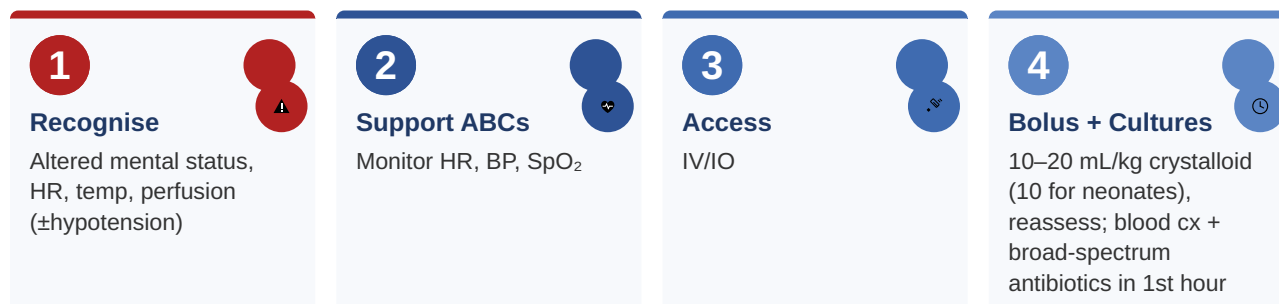
2. Classifying & Treating Shock



FLUID BOLUS

Normal saline 10–20 mL/kg IV — reassess after each bolus before repeating. Track BP (correct cuff size), urine output, ultrasound/lactate if available.

3. The 1-Hour Sepsis Bundle



4. Anaphylaxis



ADRENALINE 1:1000 (1 MG/ML, UNDILUTED)

IM 0.01 mg/kg every 15 min ×2, then every 4 hr PRN. Give without delay — do not wait for IV access.

5. Fluid Therapy — the 4Ds & ROSE

D

Drug

Fluid IS the drug

D

Dose

How much

D

Duration

How long

D

De-escalation

When to stop/remove

RESUSCITATION

Minutes · severe shock · fast, reassess · +ve balance expected



OPTIMISATION

Hours · unstable but not hypovolaemic · neutral balance



STABILISATION

Days · maintenance · normal lactate/vitals · feeds established



EVACUATION

Days–weeks · mobilise excess fluid · –ve balance targeted

Speed rule: Resuscitation boluses (10–20 mL/kg) — FAST, reassess. Rehydration/maintenance — SLOW. Sick children over-secrete ADH → restrict total intake to 50–70% of standard maintenance until feeds established.



FLUID OVERLOAD CASCADE

Excess IV fluid → interstitial leak → visceral oedema → intra-abdominal hypertension → venous hypertension → renal congestion → oliguria, low Na, SIADH. Ask every round: do I need to give fluid, or restrict/remove it?

Fluid Overload % = (Intake – Output) ÷ Baseline weight × 100. >10% is clinically significant. Track hydration, perfusion, and lactate daily.



TAKE-HOME

Isotonic fluids unless specified · crystalloids before colloids · fluid can be "poisonous" · restrict maintenance to 50–70% · Resuscitation FAST, Replacement SLOW, Maintenance SLOWER · monitor regularly.

6. Post-Op Cardiac ICU Hemodynamics

Instability peaks immediately post-op. On PICU arrival: swift, safe transfer of monitoring/infusions ("Formula 1 tyre change"), assess stability, structured handover — teamwork is everything.



WATCH FOR AT HANDOVER

Monitor/pump battery not swapped · infusion pump failure stopping critical drips · inadequate sedation causing agitation · monitor switched off mid-transfer.

First samples: ABG + ACT (ventilation, lactate, K⁺, Hb/bleeding). CXR, echo and ECG mandatory in every post-op cardiac child.

**Low Cardiac Output Syndrome**

Feeble/absent pulses, poor perfusion, rising lactate, falling urine output. Escalate before hypotension appears — it's a late sign.

**Pulmonary HTN Crisis**

After high-pulmonary-flow lesion repair: sudden desat, right heart strain, collapse. Avoid hypoxia/hypercarbia/acidosis triggers.

**Ductal-Dependent Lesions**

Keep Prostaglandin E1 (PGE1) ready and running to keep the ductus open until repair/palliation.

**LINKS TO THE HANDS-ON LAB**

Fluid-calculation workshop, timed sepsis-bundle drill, and cardiac hemodynamics case stations (low cardiac output, tamponade) — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Section 5: Management of Circulatory Problems (pp. 31–46).
- FLUID Therapy in PICU (PediCritiCon 2025 lecture) — 4Ds, ROSE principle, fluid overload monitoring.
- PeC3N — Nursing Care in the Immediate Postoperative Period, Pediatric Cardiac ICU.

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.